Transitive label budgets on six residual fixed points

Human Compression project: L044 continuation

Use the critical singleton-end setting of the accepted residual-deficit proof: $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le A_n$ is transitive, is a connected string C-group, and has every maximal omission intransitive; G_0 has orbits $\{a\}$ and its complement; $\rho_0 = (a\ b)\delta$, $K = G_{>1}$, $R = G_{>2}$ and $\sigma = \rho_1$. The odd involution δ has t pairs, with block set $\mathcal{T}$, union P, and an even fixed set F of size f in $\Omega \setminus \{a, b\}$. Assume $n = 2r \ge 14$, so K has $d = t + f/2 - 1$ labels. All block-image omission-orbit statements below use the complete accepted actual-deficit proof. Whiston's bound and the complete odd-degree maximal-string theorem, including degree five, remain paid.

Lemma 1 (Single-transposition restrictions)**.** *If K is transitive on F, $f \ge 4$, and all nonidentity distinguished restrictions to F are single transpositions, they are exactly the $f - 1$ successive edges of a natural path, each with one label.*

Proof. Their edge graph is connected. Three distinct edges incident to one vertex would have pairwise noncommuting labels, impossible in a string. A cycle is likewise impossible: labels chosen on its distinct edges would require its line graph, a cycle, to embed in the path of label positions. Hence the simple edge graph is a path.

Choose a label on each edge of that path. Their positions form a path in the integer line, so are consecutive in one of the two orders. Any repeated label on an interior edge would have to be adjacent to both chosen neighboring labels, whose unique common neighbor is the already occupied position. For an end edge, its neighboring edge is interior because $f \ge 4$; both positions adjacent to that neighboring label are already occupied. A second end-edge label is also impossible. □

Lemma 2 (Minimum transitive fixed-set budget)**.** *Let E be a set of h distinguished K labels whose restrictions are transitive on F. Then $h \ge f/2$. If equality holds and $t \ge 3$ is odd, the other $t-1$ pair images generate the ordinary S_t path, at consecutive positions J, and every label of E has trivial pair image.*

Proof. Deleting a label outside E leaves the fixed-set action transitive. Its pair image cannot be generated by the other pair images, as that would leave the full block-orbit partition unchanged. In particular the $d - h$ pair images outside E are independent. Whiston gives $d - h \le t - 1$, or $h \ge f/2$.

At equality those $t - 1$ images generate S_t and form its ordinary path by the accepted odd-degree maximal-string theorem. For $t = 3$ this just says that two independent involutions in S_3 are distinct transpositions. Consecutive noncommutations force consecutive positions in the original string. Omitting any path mark leaves two pair orbits. If an E label joined them, the full omission would be transitive both on pairs and on F, contradicting the actual orbit deficit. Thus every E label preserves every path cut; their common stabilizer fixes every pair-block. This proves the last assertion. □

Lemma 3 (The first label cannot be an extra)**.** *Let E consist of $h = f/2 \ge 2$ distinguished labels, transitive on F, and suppose $t \ge 3$ is odd. Then $\rho_2 \notin E$.*

Proof. Suppose $\rho_2 \in E$. Then R contains all of J, so is transitive on $\mathcal{T}$ and fixes no point of P. Because K fixes b, transitivity of $G_0 = \langle \sigma, K \rangle$ forces $\sigma(b) \neq b$. Also $\sigma(b) \neq a$, since σ fixes a. Centralization shows that $z = \sigma(b)$ is fixed by R; hence $z \in F$.

Partition the positions of E into maximal intervals. Distinct intervals commute, by the string relations. Any interval not containing ρ_2 is contained in R, and its subgroup fixes z. It also commutes with the subgroup on all the other intervals. Consequently it fixes the entire $\langle E \rangle$-orbit of z: each orbit point can be written z^u with u in those other intervals, because the interval itself fixes z. This orbit is F, by transitivity. Removing even one label of that interval would then leave the same transitive action on F, contradicting the minimum h-label budget.

There is therefore only one E interval, beginning at ρ_2. All J labels follow it. Since $|J| = t - 1 \geq 2$, its last label λ is at distance at least two from every E label, and centralizes $\langle E \rangle$. It belongs to R, so fixes z, and hence fixes all of F pointwise. It also fixes a, b.

On the other hand, λ acts nontrivially on every R-orbit $O \subseteq P$. Such an orbit meets every pair-block, since R is transitive on $\mathcal{T}$; the nontrivial pair transposition of λ therefore moves points of O. The permutation σ transports R-orbits equivariantly. No such O can be sent outside P, where λ fixes every point. Thus σ preserves P. So does K, contradicting transitivity on the complementary G_0 orbit. This proves the lemma. □

Theorem 4 (The strict transitive label budget)**.** *Suppose $f \geq 4$ and $t \geq 3$. Every transitive fixed-set subfamily has at least $f/2 + 1$ labels. In particular no three distinguished labels can be transitive on a six-point fixed set, including $t = 3$, $n = 14$.*

Proof. Lemma 2 gives $h \geq f/2$; suppose equality holds. By Lemma 3, ρ_2 belongs to J. Its consecutive path therefore occupies the initial positions, and all E labels lie in R. Thus F is an R-orbit. If σ preserved it, then G_0 would preserve it too. Hence $\sigma(F) = O \subseteq P$; the singleton orbits $\{a\}, \{b\}$ cannot be its image.

Equivariance makes $\langle E \rangle$ transitive on O. But every E label has trivial pair image: each individual pair-block is invariant under $\langle E \rangle$. Its orbits in P consequently have size at most two, contradicting $|O| = |F| = f \geq 4$. Equality is impossible. □

Exact six-point consequence. A distinguished triple transposition on a transitive six-point F can be extended from its three disjoint edges to a spanning tree using two further edges. At most three labels are then transitive on F, contrary to Theorem 4 and the minimum budget. Thus no triple transposition occurs. If every nonidentity restriction is a single transposition, the separate path-label lemma and natural fixed-path exclusion apply. The remaining transitive six-point case has at least one double transposition and minimum transitive label number exactly four: a spanning tree extending its two edges uses at most three further labels, while the lower bound and Theorem 4 exclude three. This last case is not resolved here.

Attribution and boundary. The source is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, physical pages38–44. The path-label observation and the accepted four-fixed argument of collaborator `rewrite_termination` supply the underlying equivariant-orbit method; the full preceding actual-deficit argument remains paid. Lemmas 1–2 are unchanged from the preceding label-budget edition; its weaker three-label theorem is strengthened here. The fixed point of R, interval-minimality and distant path-label obstruction are the present continuation. The final individual-pair orbit observation was also explicitly supplied by the root collaborator while this successor was being completed. No numerical search, projected IP, smaller-degree alternating induction or new formal theorem is used. The intransitive six-point actions, four-label case, larger fixed sets and other principal L044 branches remain open in this recovery. No whole-original or question credit is claimed.